Atliq Hardware

About Atliq

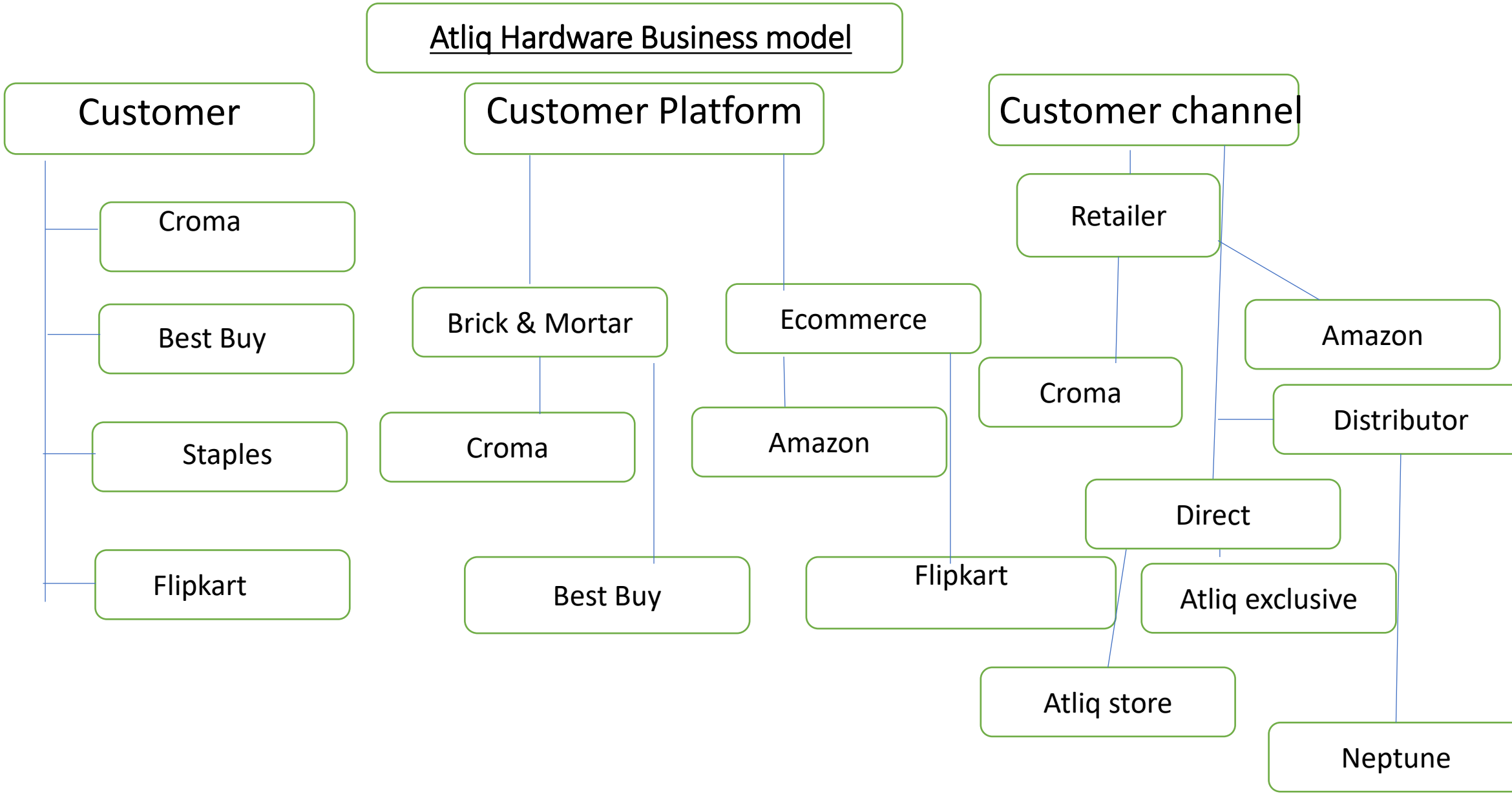
Atliq is a hardware company specializing in pc, printers , mouse and computers with global reach

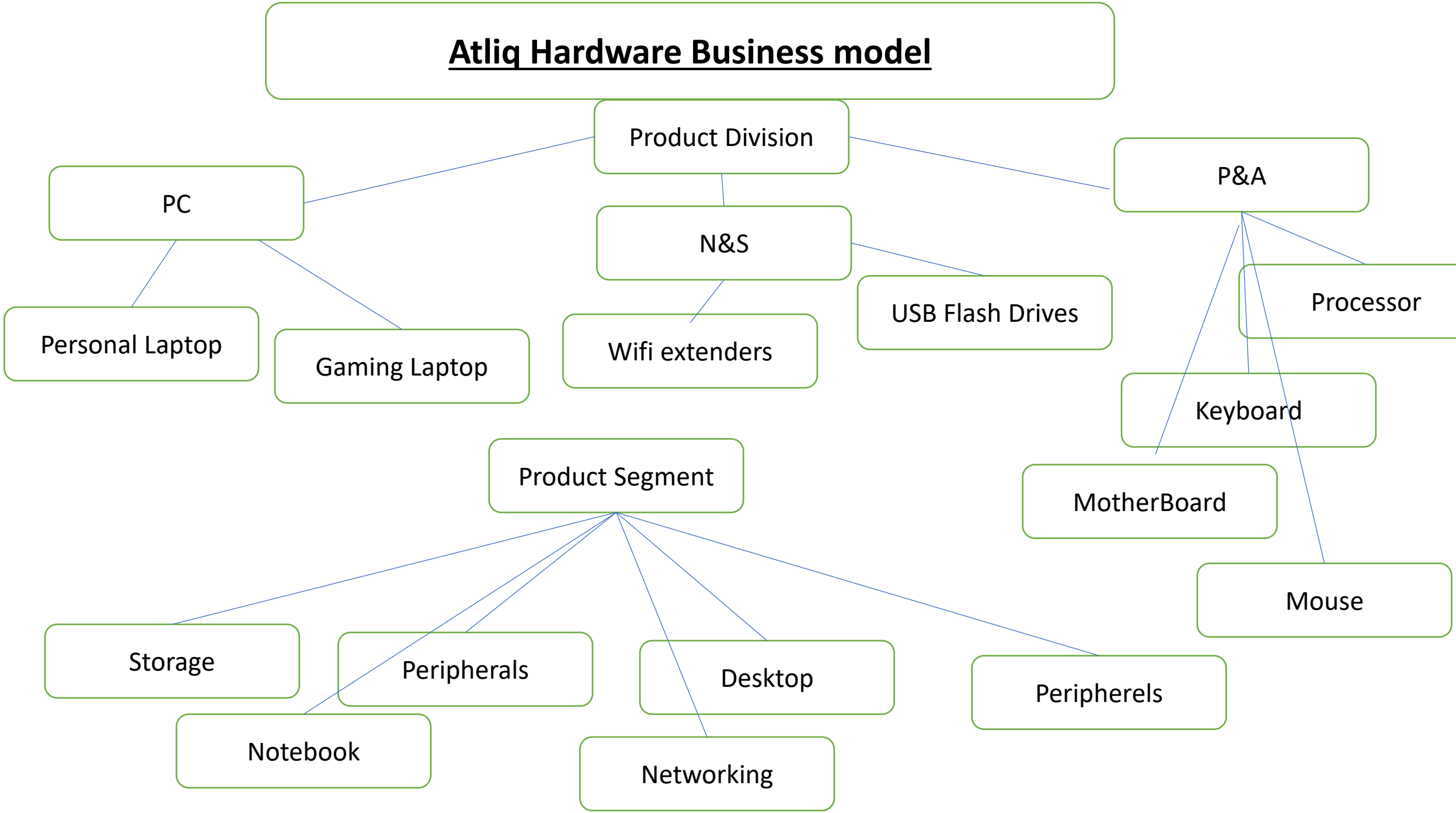
Problem Statement

They faced major problem in their business for it they need quick and data informed decisions to stay competitive in the competitive market management also noticed they missing critical insights for strategic moves

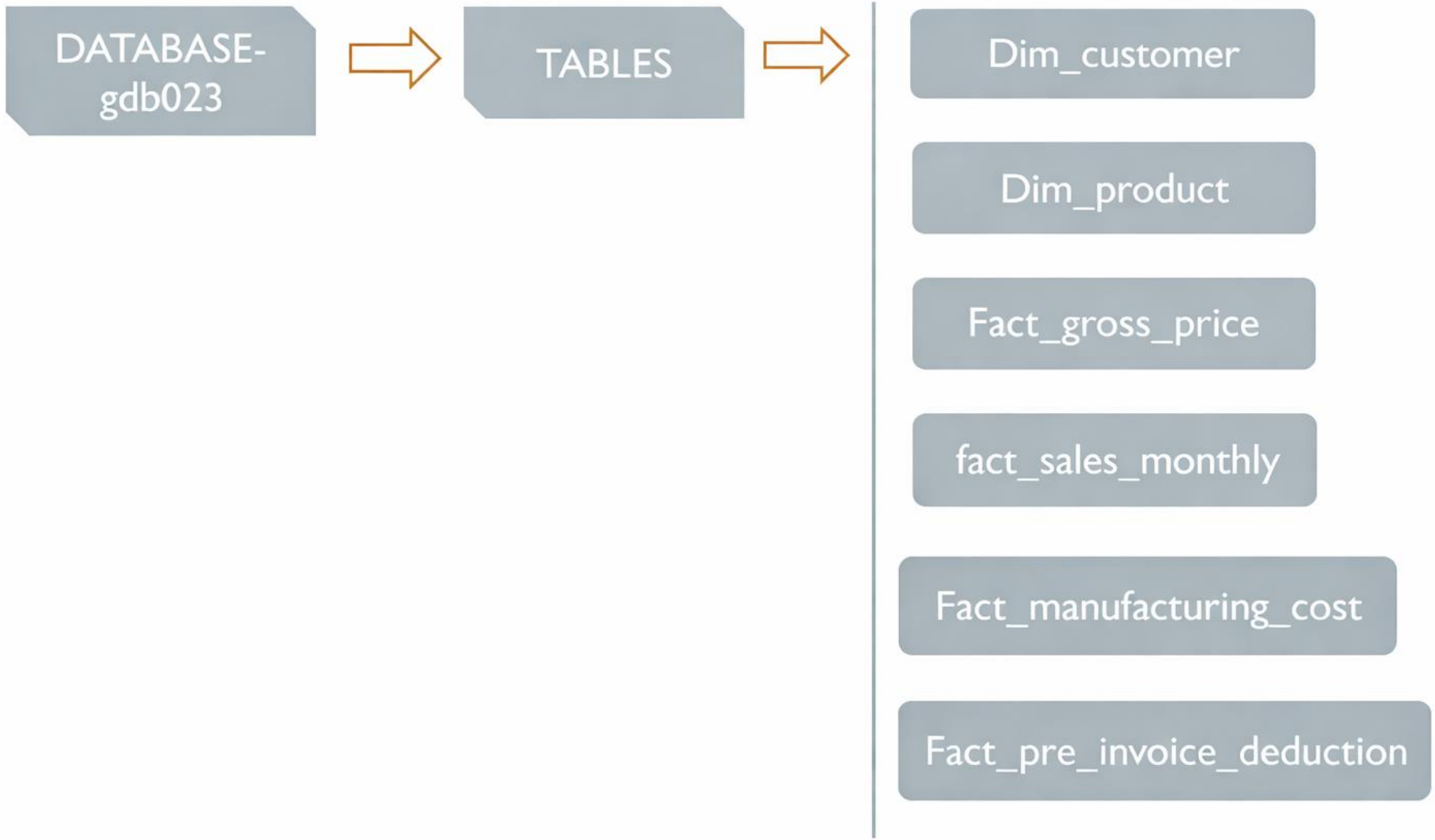
Project overview

In this project I will be working with consumer goods dataset related to consumer goods. The goal is to answer 10 specific sql ad hoc request


Atliq Hardware Business model
Customer
Croma
Best Buy
Staples
Flipkart
Customer Platform
Brick & Mortar
Croma
Best Buy
Ecommerce
Amazon
Flipkart
Customer channel
Retailer
Croma
Amazon
Distributor
Direct
Atliq exclusive
Atliq store
Neptune


Atliq Hardware Business model
Product Division
PC
N&S
P&A
Personal Laptop
Gaming Laptop
Wifi extenders
USB Flash Drives
Processor
Keyboard
Product Segment
MotherBoard
Mouse
Storage
Peripherals
Desktop
Peripherels
Notebook
Networking

DATA-SET


DATABASE-
gdb023
TABLES
Dim_customer
Dim_product
Fact_gross_price
fact_sales_monthly
Fact_manufacturing_cost
Fact_pre_invoice_deduction

Ad_Hoc request1

Question

Provide the list of markets in which customer "Atliq Exclusive" operates its business in the APAC region

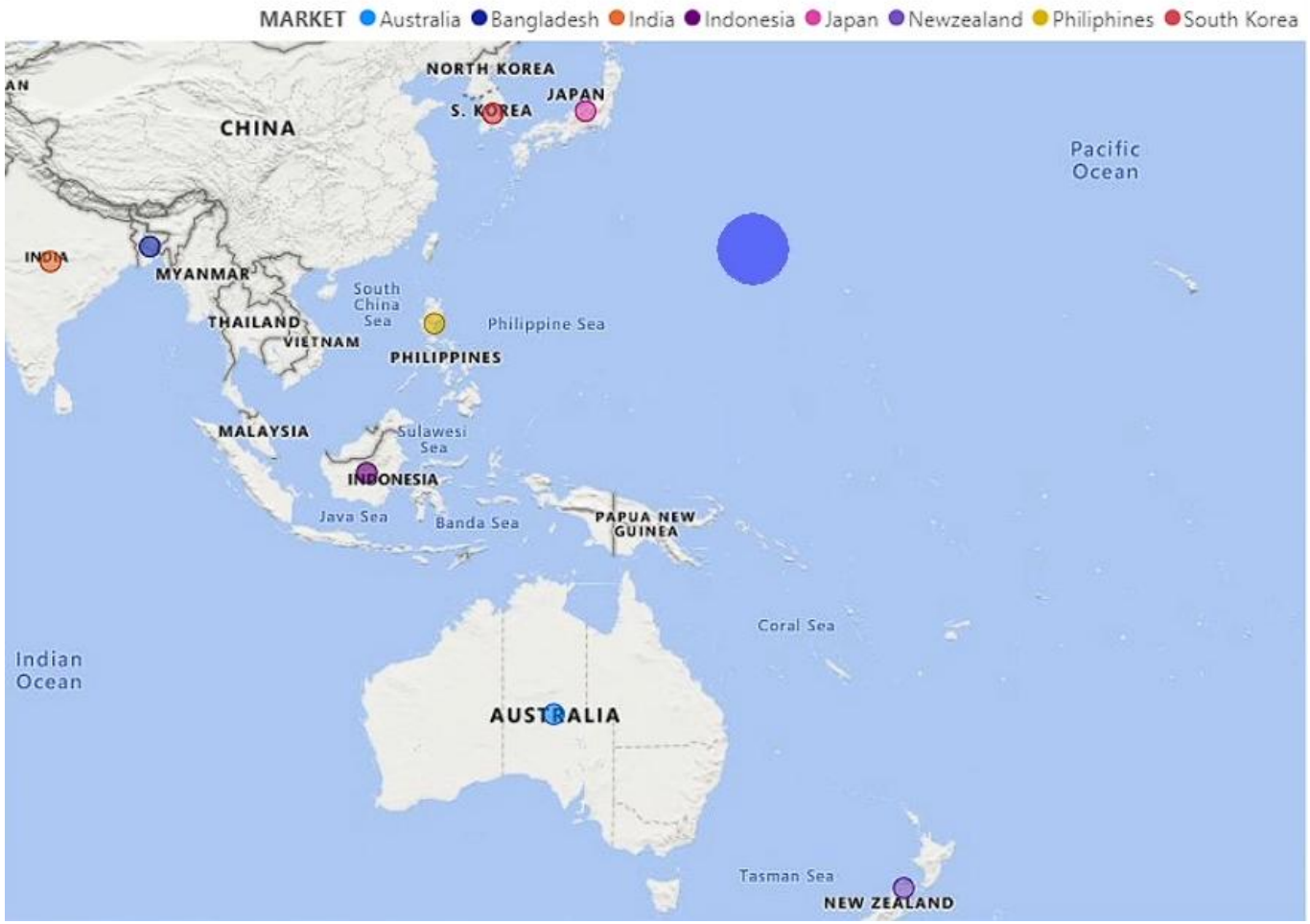
Input

```sql
SELECT
    market
FROM
    dim_customer
WHERE
    customer = 'Atliq Exclusive'
        AND region = 'APAC'
```

Output



Visual



Insights1

- **"Atliq Exclusive"** has a presence in several countries across the **APAC region**, including **India, Indonesia, Japan, the Philippines, South Korea, Australia, New Zealand, and Bangladesh.**
- It reflects a **robust market presence** and **adaptability** to different cultural and economic contexts in the Asia-Pacific region.

Ad_Hoc request2

Question

"What is the percentage of unique product increase in 2021 vs. 2020? The final output contains these fields, unique_products_2020 unique_products_2021 percentage_chg"

Input

```sql
WITH yearly_unique_products AS (
    SELECT
        fiscal_year,
        COUNT(DISTINCT product_code) AS unique_products
    FROM fact_sales_monthly
    WHERE fiscal_year IN (2020, 2021)
    GROUP BY fiscal_year)

SELECT
    MAX(CASE WHEN fiscal_year = 2020 THEN unique_products END)
        AS unique_products_2020,

    MAX(CASE WHEN fiscal_year = 2021 THEN unique_products END)
        AS unique_products_2021,

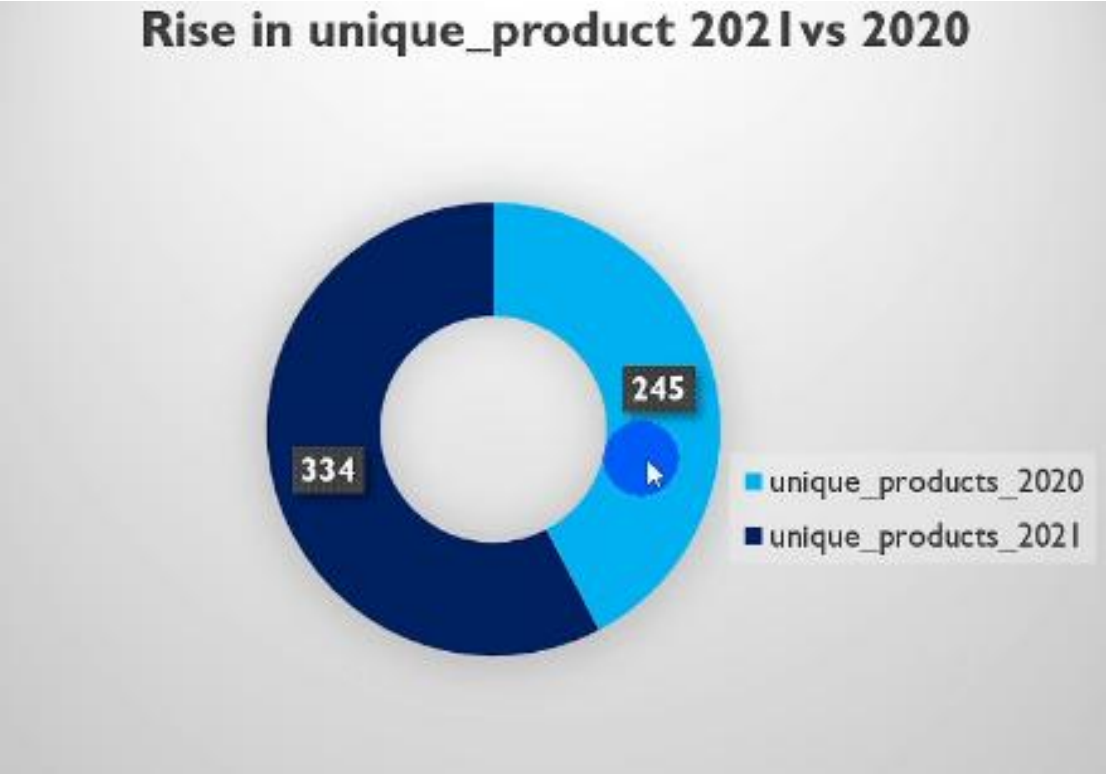
    ROUND(
        (
            MAX(CASE WHEN fiscal_year = 2021 THEN unique_products END)
            -
            MAX(CASE WHEN fiscal_year = 2020 THEN unique_products END)
        ) * 100.0
        /
        MAX(CASE WHEN fiscal_year = 2020 THEN unique_products END)
    , 2) AS percentage_change
FROM yearly_unique_products;
```

Output

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

unique_products_2020	unique_products_2021	percentage_change
245	334	36.33

Visual2



Insights

- **Significant Growth**: There was a significant increase in unique products, with **334 in 2021** compared to **245 in 2020**.
- **Percentage Increase**: The percentage change represents a **growth of 36.33%** in unique products from one year to the next.
- **Strategic Focus**: This substantial increase in unique products suggests a focus on **expanding product offerings**, which can attract a broader customer base and potentially boost sales and revenue.
- **Business Health**: It's a positive indicator of **business growth** and **adaptability** to changing market demands.

Ad_Hoc request3

Question

"Provide a report with all the unique product counts for each segment and sort them in descending order of product counts. The final output contains 2 fields, segment product_count"

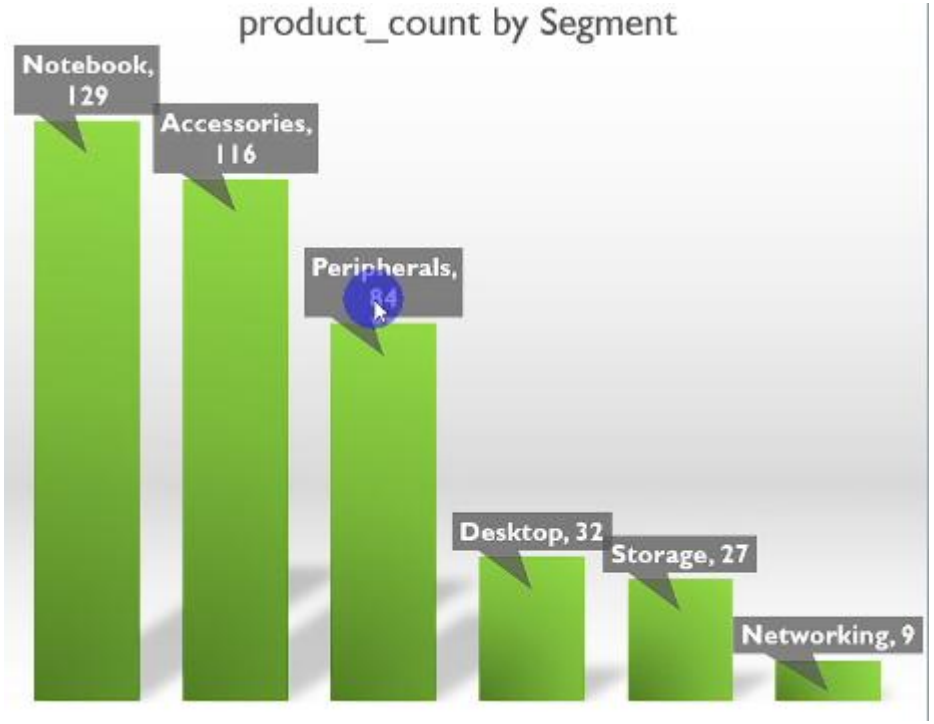
Input

```
1 • SELECT segment, count(distinct(product_code)) as product_count
2   from dim_product
3   group by segment
4   order by product_count desc;
```

Output

segment	product_count
Notebook	129
Accessories	116
Peripherals	84
Desktop	32
Storage	27
Networking	9

Visual



Insights 3

- The **"Notebook"** segment has the highest product count, with **129 products**.
- The **"Networking"** segment has the smallest product count with only **9 products**.
- These insights indicate the diversity and product offerings within each segment, with "Notebook" and "Accessories" offering a wide range of options, while "Networking" has a more limited selection.
- The variety in product offerings allows catering to various customer preferences and needs across different segments.

Ad_Hoc request4

Question

Which segment had the most increase in unique products in 2021 vs 2020? The final output contains these fields, segment product_count_2020 product_count_2021 difference"

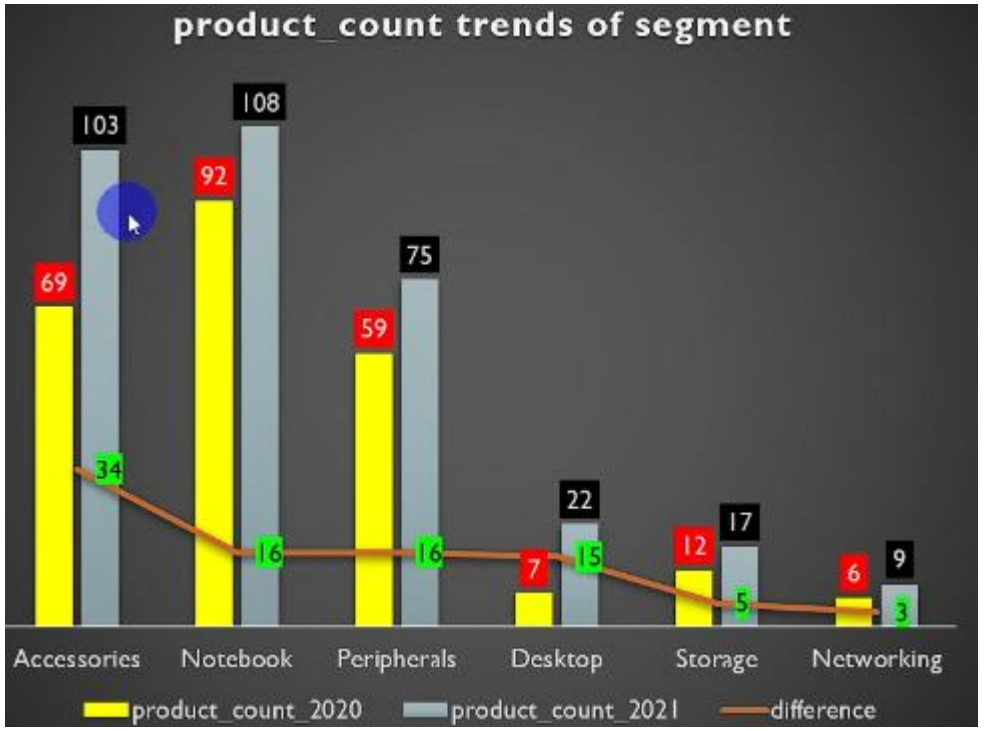
Input

```
with x as (
    select p.segment,
           count(distinct s.product_code) as product_count_2020
    from dim_product p
    join fact_sales_monthly s
         on p.product_code = s.product_code
    where s.fiscal_year = 2020
    group by p.segment
),
y as (
    select p.segment,
           count(distinct s.product_code) as product_count_2021
    from dim_product p
    join fact_sales_monthly s
         on p.product_code = s.product_code
    where s.fiscal_year = 2021
    group by p.segment
)
 select x.segment,
        x.product_count_2020,
        y.product_count_2021,
        (y.product_count_2021 - x.product_count_2020) as difference
 from x
 join y on x.segment = y.segment
 order by difference desc;
```

Output

segment	product_count_2020	product_count_2021	difference
Accessories	69	103	34
Notebook	92	108	16
Peripherals	59	75	16
Desktop	7	22	15
Storage	12	17	5
Networking	6	9	3

Visual



Insights

- The **"Accessories"** segment saw a significant increase in product counts, with **34 more products** in 2021 compared to 2020.
- These insights suggest a general trend of **product portfolio growth**, particularly in the "Accessories" and "Notebook" segments, which can indicate a strategy to cater to a **broader range of customer preferences**.
- The growth in product variety may **enhance market competitiveness** and offer customers more choices.

Ad_Hoc request5

Question

"Get the products that have the highest and lowest manufacturing costs. The final output should contain these fields, product_code product manufacturing_cost"

Input

```sql
SELECT
    fmc.product_code,
    dp.product,
    fmc.manufacturing_cost
FROM fact_manufacturing_cost fmc
JOIN dim_product dp
    ON fmc.product_code = dp.product_code
WHERE fmc.manufacturing_cost = (
        SELECT MAX(manufacturing_cost)
        FROM fact_manufacturing_cost
    )
   OR fmc.manufacturing_cost = (
        SELECT MIN(manufacturing_cost)
        FROM fact_manufacturing_cost
    )
   OR fmc.manufacturing_cost = (
        SELECT MIN(manufacturing_cost)
        FROM fact_manufacturing_cost
    );
```

Insight

Output

product_code	product	manufacturing_cost
A2118150101	AQ Master wired x1 Ms	0.8920
A6120110206	AQ HOME Allin1 Gen 2	240.5364

- **"AQ HOME Allin1 Gen 2"** has a relatively higher manufacturing cost of **240.5364**.
- In contrast, **"AQ Master wired x1 Ms"** has a significantly lower manufacturing cost of **0.892**.

Ad_Hoc request6

Question

"Generate a report which contains the **top 5 customers** who received an **average high pre_invoice_discount_pct** for the **fiscal year 2021** and in the **Indian market**. The final output contains these fields, **customer_code**, **customer**, **average_discount_percentage**"

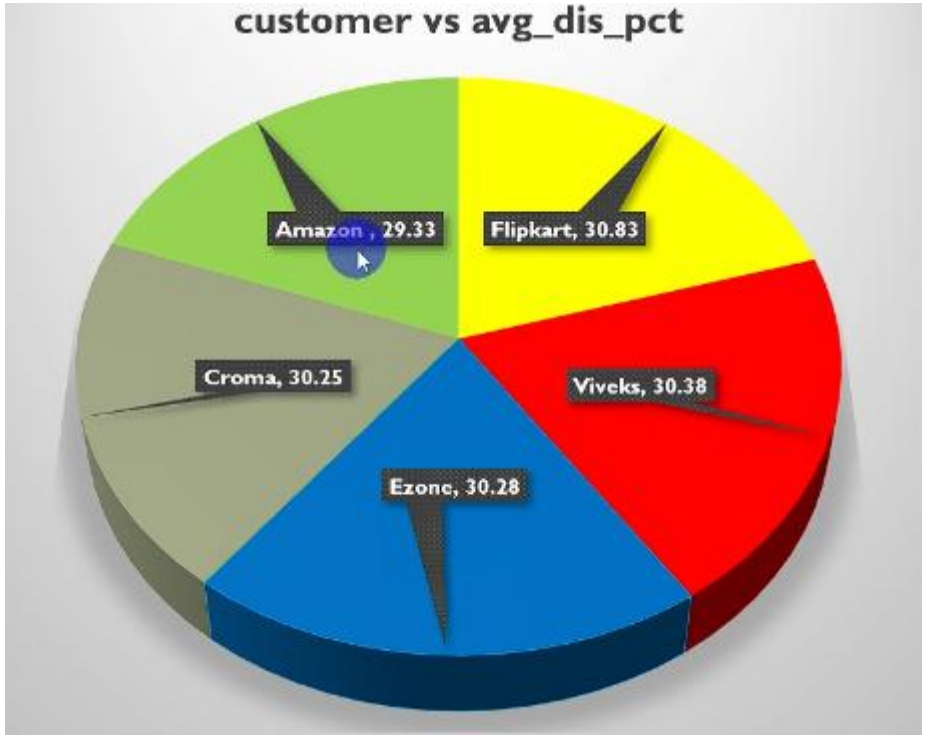
Input

```sql
SELECT
    c.customer_code,
    c.customer AS customer,
    ROUND(AVG(pid.pre_invoice_discount_pct) * 100, 2)
        AS average_discount_percentage
FROM fact_pre_invoice_deductions pid
JOIN dim_customer c
    ON pid.customer_code = c.customer_code
WHERE pid.fiscal_year = 2021
  AND c.market = 'India'
GROUP BY
    c.customer_code,
    c.customer_code,
    c.customer
 ORDER BY
    average_discount_percentage DESC
 LIMIT 5;
```

Output

customer_code	customer	average_discount_perce
90002009	Flipkart	30.83
90002006	Viveks	30.38
90002003	Ezone	30.28
90002002	Croma	30.25
90002016	Amazon	29.33

Visual



Insights

- **"Flipkart"** has the highest average pre-invoice discount percentage at **30.83%**.
- **"Amazon"** has a comparatively lower average discount at **29.33%**.
- These insights indicate **varying discount strategies** among customers, with "Flipkart" and "Viveks" offering the highest average discounts, potentially attracting **cost-conscious shoppers**.
- **"Amazon"** provides relatively lower discounts, suggesting a different pricing strategy or a customer base **less sensitive to discounts**.
- These insights can be valuable for **adjusting discount strategies** and understanding customer preferences.

Ad_Hoc request7

Question

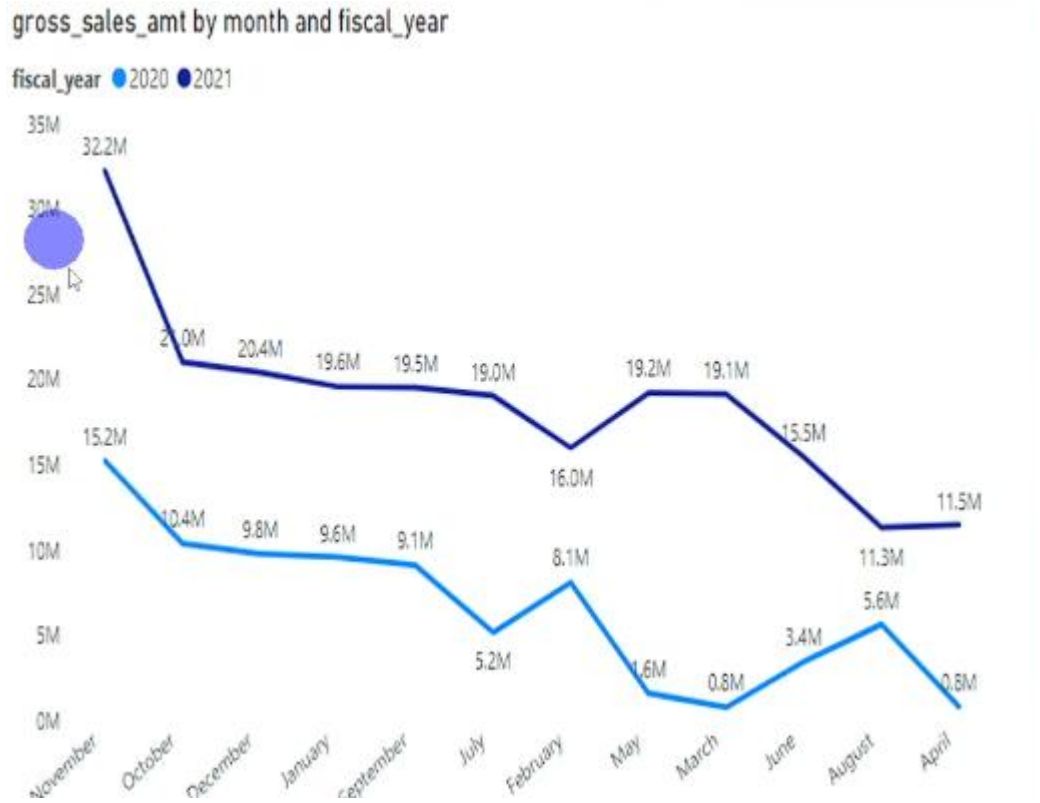
"Get the complete report of the **Gross sales amount** for the customer **'Atliq Exclusive'** for **each month**. This analysis helps to get an idea of low and high-performing months and take strategic decisions. The final report contains these columns: **Month**, **Year**, **Gross sales Amount**."

Input

```sql
SELECT
    MONTHNAME(s.date) AS month,
    s.fiscal_year,
    ROUND(SUM(g.gross_price * s.sold_quantity), 2) AS gross_sales_amt
FROM fact_sales_monthly s
JOIN dim_customer c
    ON s.customer_code = c.customer_code
JOIN fact_gross_price g
    ON s.product_code = g.product_code
WHERE c.customer = 'atliq exclusive'
GROUP BY MONTHNAME(s.date), s.fiscal_year
ORDER BY s.fiscal_year;
```

month	fiscal_year	gross_sales_amt
September	2020	9092670.34
October	2020	10378637.60
November	2020	15231894.97
December	2020	9755795.06
January	2020	9584951.94
February	2020	8083995.55
March	2020	766976.45
April	2020	800071.95
May	2020	1586964.48
June	2020	3429736.57
July	2020	5151815.40
August	2020	5638281.83
September	2021	19530271.30
October	2021	21016218.21
November	2021	32247289.79
December	2021	20409063.18
January	2021	19570701.71
February	2021	15986603.89
March	2021	19149624.92
April	2021	11483530.30
May	2021	19204309.41
June	2021	15457579.66
July	2021	19044968.82
August	2021	11324548.34

Visual



Insight

- **November 2021** had the highest gross sales amount, reaching **$32,247,289.79**.
- In contrast, the fiscal year 2021 started with lower sales in **September** but still had a significant peak in November.
- There is a **notable seasonality** in sales, with November being a consistently strong month.
- The months of **March and April** in fiscal year 2020 had relatively low sales, which improved in fiscal year 2021.
- These insights can guide **strategic decisions**, such as focusing marketing efforts and inventory planning around the peak sales months and addressing potential challenges during lower sales months

Ad_Hoc request8

Question

"In which **quarter of 2020**, got the **maximum total_sold_quantity**? The final output contains these fields sorted by the **total_sold_quantity**: **Quarter**, **total_sold_quantity**".

Input

```
1  SELECT
2      CASE
3          WHEN MONTH(date) IN (9, 10, 11) THEN 'Q1'
4          WHEN MONTH(date) IN (12, 01, 02) THEN 'Q2'
5          WHEN MONTH(date) IN (03, 04, 05) THEN 'Q3'
6          ELSE 'Q4'
7      END AS Quarters,
8      SUM(sold_quantity) AS total_sold_qty
9  FROM fact_sales_monthly
10 WHERE fiscal_year = 2020
11 GROUP BY Quarters
12 ORDER BY total_sold_qty DESC;
```

Output

Result Grid | Filter Rows:

Quarters	total_sold_qty
Q1	7005619
Q2	6649642
Q4	5042541
Q3	2075087

Visual



Insight

- The highest total sold quantity is in **Q1**, with **7,005,619 units**.
- These insights highlight a seasonal variation in sales, with **Q1 and Q2** being the strongest quarters and **Q3** being the weakest.
- This information is valuable for planning inventory and marketing strategies to align with seasonal demand.

Ad_Hoc request9

Question

"Which **channel** helped to bring more gross sales in the **fiscal year 2021** and the **percentage of contribution**? The final output contains these fields, **channel**, **gross_sales_mln**, **percentage**".

Input

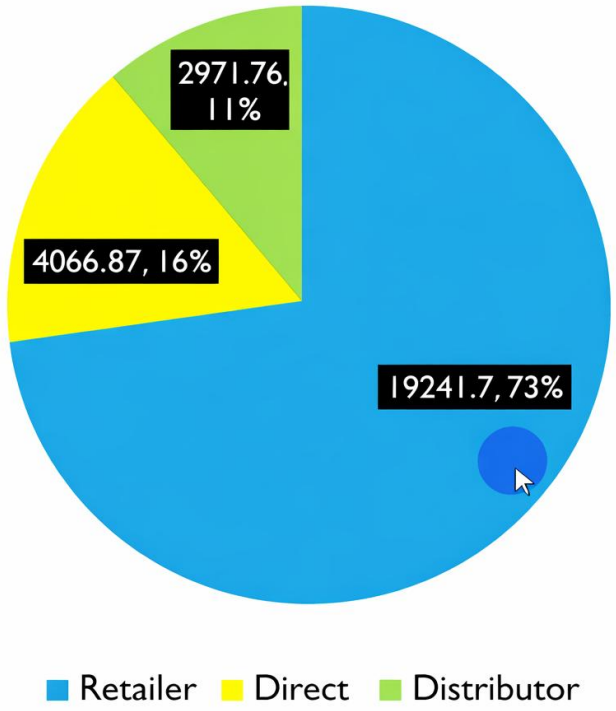
```sql
WITH x AS (
    SELECT
        c.channel,
        ROUND(SUM(g.gross_price * s.sold_quantity) / 1000000, 2) AS gross_sales_mln
    FROM fact_sales_monthly s
    JOIN dim_customer c USING (customer_code)
    JOIN fact_gross_price g USING (product_code)
    WHERE s.fiscal_year = 2021
    GROUP BY c.channel
)

SELECT
    channel,
    gross_sales_mln,
    ROUND(
    ROUND(
        (gross_sales_mln / (SELECT SUM(gross_sales_mln) FROM x)) * 100,
        2
    ) AS pct
FROM x
ORDER BY gross_sales_mln DESC;
ORDER BY total sold qty DESC;
```

Output

channel	gross_sales_mln	pct
Retailer	1924.17	73.22
Direct	406.69	15.48
Distributor	297.18	11.31

Visual



Insight

- The "Retailer" channel accounts for the majority of sales, contributing to 73.22% of gross sales.
- The "Direct" channel also plays a significant role, representing 15.47% of gross sales
- The "Distributor" channel contributes 11.31% of gross sales.
- A significant focus on the "Retailer" channel suggests it is the primary revenue driver.
- Diversification and growth opportunities may be explored in the "Direct" and "Distributor" channels to further maximize sales.

Ad_Hoc request10

Question

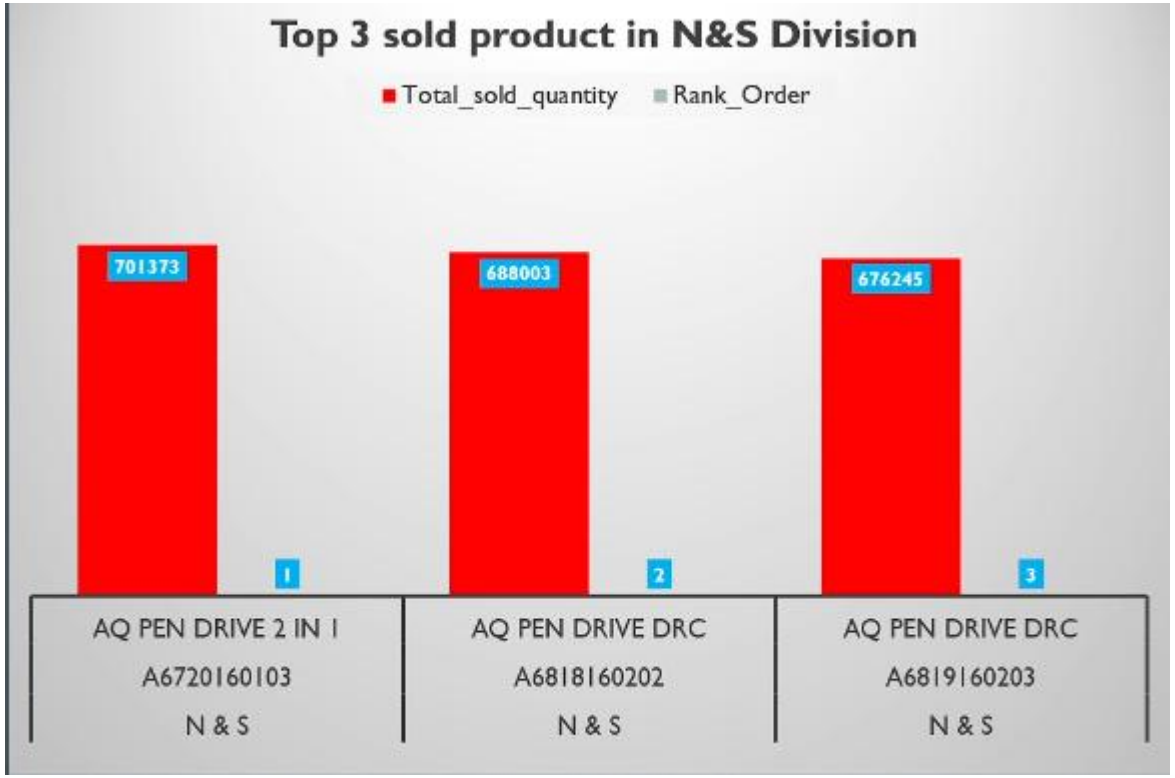
"Get the **Top 3 products** in each division that have a high **total_sold_quantity** in the **fiscal_year 2021**? The final output contains these fields, **division**, **product_code**".

```
WITH  x AS
(
SELECT P.division, S.product_code, P.product, SUM(S.sold_quantity) AS Total_sold_quantity,
 RANK() OVER(PARTITION BY P.division ORDER BY SUM(S.sold_quantity)  DESC) AS 'Rank_Order'
FROM dim_product P JOIN fact_sales_monthly S
ON P.product_code = S.product_code
WHERE S.fiscal_year = 2021
GROUP BY  P.division, S.product_code,  P.product)

 SELECT *  FROM x
WHERE  Rank_Order IN (1,2,3) ORDER BY division,  Rank_Order;
```

division	product_code	product	Total_sold_quantity	Rank_Order
N & S	A6720160103	AQ Pen Drive 2 IN 1	701373	1
N & S	A6818160202	AQ Pen Drive DRC	688003	2
N & S	A6819160203	AQ Pen Drive DRC	676245	3
P & A	A2319150302	AQ Gamers Ms	428498	1
P & A	A2520150501	AQ Maxima Ms	419865	2
P & A	A2520150504	AQ Maxima Ms	419471	3
PC	A4218110202	AQ Digit	17434	1
PC	A4319110306	AQ Velocity	17280	2
PC	A4218110208	AQ Digit	17275	3

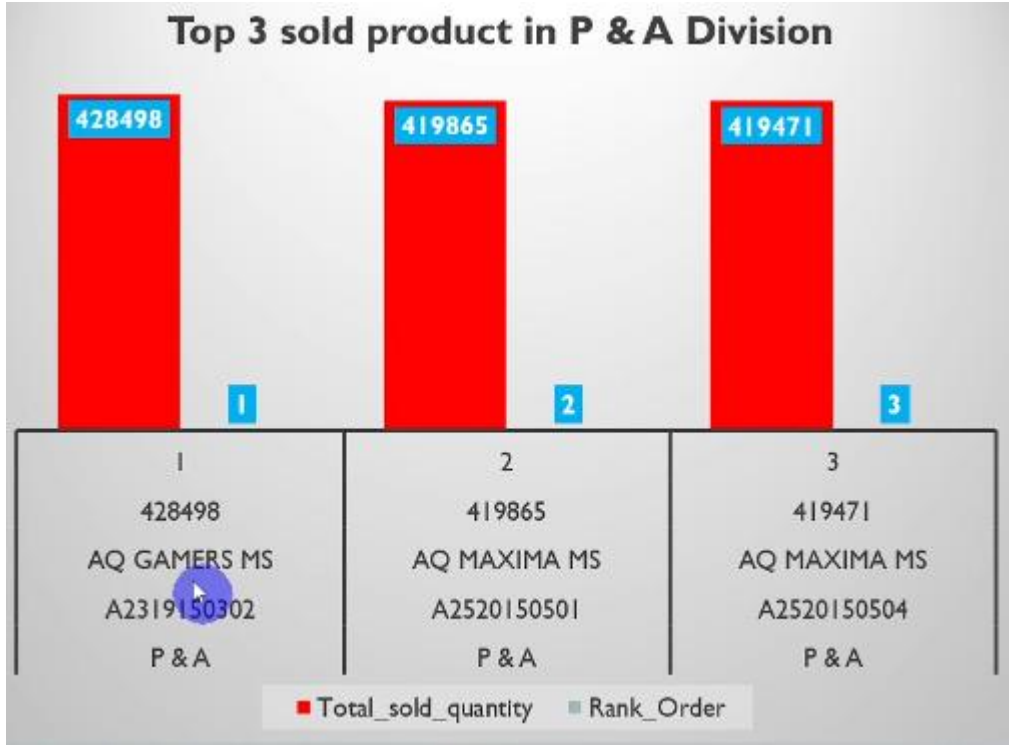
Visual



Insight

- The top three products: "AQ Pen Drive 2 IN 1," "AQ Pen Drive DRC," and another variant of "AQ Pen Drive DRC.“
- Strong consumer preference for pen drives and related items in this division
- The division's leadership in this market segment, indicating growth potential
- Efficient inventory management is crucial for sustained success.

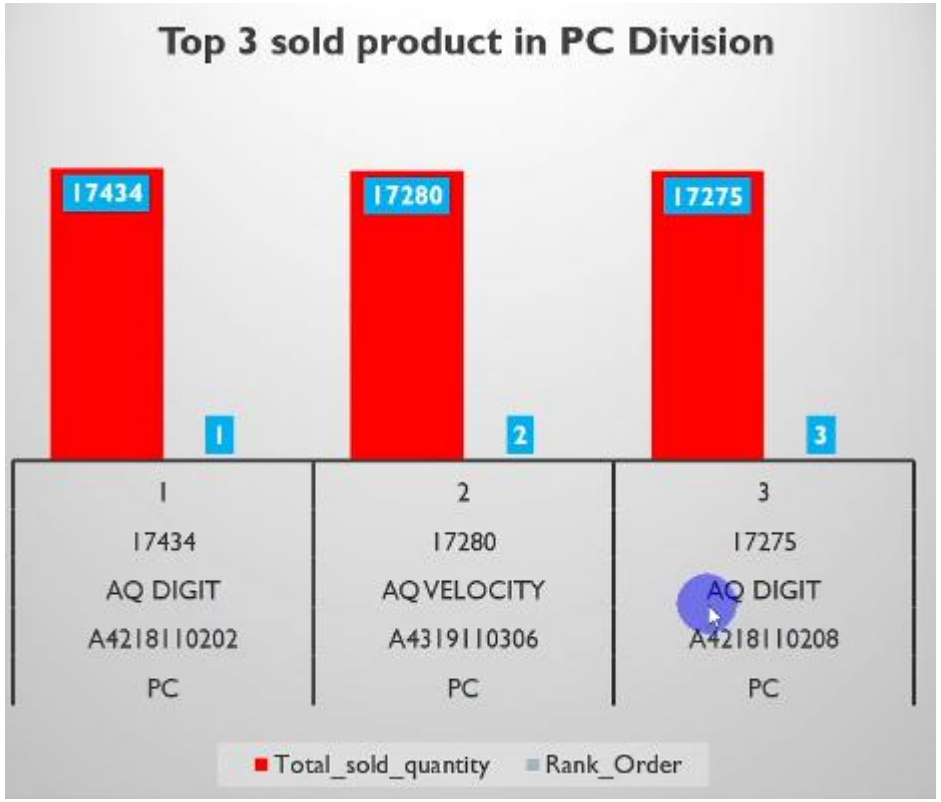
Visual



Insight

- The best-selling product is "AQ Gamers Ms," followed by two variants of "AQ Maxima Ms.“
- This division specializes in gaming-related products, appealing to a specific customer niche.
- Efficient inventory management is essential to meet customer demand.
- Continuously promote and brand the top products to sustain success.

Visual



Insight

- The best-selling product in the "PC" division is "AQ Digit," with two variants followed by "AQ velocity".
- The division specializes in PC-related products, indicating a focus on the computing niche
- Efficient inventory management is essential to meet customer demand.
- Ongoing branding and marketing efforts are important to maintain success.